AUV / ROV SHIP HULL STRUCTURAL INSPECTION REPORT

Autonomous Robotic NDT Survey • Generated: 2026-10-17 15:10:54 • Mission ID: INSP_20261017_151015_5dd76d

1. Executive Mission Summary

Metric	Value	Metric	Value
Inspection ID	INSP_20261017_151015_5dd76d	Start Timestamp	2026-10-17T15:10:16
Hull Surface Inspected	14.85%	Total Logged Steps	120
Suspected Defects Found	120	Mean Detection Confidence	100.0%
Water Visibility Range	0.75 - 0.75	ROV Battery Remaining	99.0%

2. Suspected Hull Defects Detected

Step	Zone ID	Pose (X, Y) [m]	Defect Category	Confidence	Inspection Status
1	Bow-Keel-Bottom	(1.0, 0.6)	Corrosion	100.0%	Requires Inspection
2	Bow-Keel-Bottom	(1.4, 0.6)	Corrosion	100.0%	Requires Inspection
3	Bow-Keel-Bottom	(1.8, 0.6)	Corrosion	100.0%	Requires Inspection
4	Bow-Keel-Bottom	(2.2, 0.6)	Corrosion	100.0%	Requires Inspection
5	Bow-Keel-Bottom	(2.6, 0.6)	Corrosion	100.0%	Requires Inspection
6	Bow-Keel-Bottom	(3.0, 0.6)	Corrosion	100.0%	Requires Inspection
7	Bow-Keel-Bottom	(3.4, 0.6)	Corrosion	100.0%	Requires Inspection
8	Bow-Keel-Bottom	(3.8, 0.6)	Corrosion	100.0%	Requires Inspection
9	Bow-Keel-Bottom	(4.2, 0.6)	Corrosion	100.0%	Requires Inspection
10	Bow-Keel-Bottom	(4.6, 0.6)	Corrosion	100.0%	Requires Inspection
11	Bow-Keel-Bottom	(5.0, 0.6)	Corrosion	100.0%	Requires Inspection
12	Bow-Keel-Bottom	(5.4, 0.6)	Corrosion	100.0%	Requires Inspection
13	Bow-Keel-Bottom	(5.8, 0.6)	Corrosion	100.0%	Requires Inspection
14	Bow-Keel-Bottom	(6.2, 0.6)	Corrosion	100.0%	Requires Inspection
15	Bow-Keel-Bottom	(6.6, 0.6)	Corrosion	100.0%	Requires Inspection
16	Bow-Keel-Bottom	(7.0, 0.6)	Corrosion	100.0%	Requires Inspection
17	Bow-Keel-Bottom	(7.4, 0.6)	Corrosion	100.0%	Requires Inspection
18	Bow-Keel-Bottom	(7.8, 0.6)	Corrosion	100.0%	Requires Inspection
19	Bow-Keel-Bottom	(8.2, 0.6)	Corrosion	100.0%	Requires Inspection
20	Bow-Keel-Bottom	(8.6, 0.6)	Corrosion	100.0%	Requires Inspection

3. Scientific Limitations & Marine Engineering Disclaimer

SCIENTIFIC & ENGINEERING LIMITATIONS:

1. This automated inspection system identifies visible surface corrosion and cracking patterns using optical computer vision. Internal metallurgical stress, micro-fatigue, and subsurface cracks cannot be measured from RGB camera imagery alone.
2. All model detections are classified as *'suspected defects'* with status *'requires further inspection'*. Confidence scores represent statistical model activation, not calibrated structural failure probabilities.
3. Confirmatory structural integrity assessment requires certified marine surveyor evaluation and complementary non-destructive testing (NDT), such as ultrasonic thickness gauging (UT) or eddy current testing.